Residual Gas Velocity Above a Stopped Propeller Blade in Dense Xenon: A Mechanism-by-Mechanism Analysis

1 Problem Definition

A horizontal two-bladed propeller with NACA 0012 airfoil cross-section rotates inside a vertical cylindrical vessel filled with xenon gas at high pressure. The propeller sweeps through a prescribed angle θ_{sweep} in a time T_{sweep}, starting from rest and returning to rest. We seek the residual gas velocity at a height z_{obs} above the upper blade surface, evaluated at a time t_{obs} after the blade has stopped.

1.1 Parameters

Seven parameters are treated as variable inputs; all others are fixed. The rotation time t_{rot} is computed from the inertia and motor torque (see Section 3).

Table 1: Variable parameters (with default values).

Symbol	Description	Default	Units
c	Blade chord length	0.16	m
θ_{sweep}	Sweep angle	π	rad
m_{blade}	Mass of each blade ($\times 2$)	0.25	kg
m_{plate}	Mass of each carrier plate ($\times 2$)	0.15	kg
τ	Motor torque	60	N·m
z_{obs}	Observation height above blade top	0.005	m
t_{obs}	Observation time after stop	0.5	s

Table 2: Fixed parameters.

Parameter	Value	Description
R	1.6 m	Blade span (hub to tip)
P	15 bar	Xenon pressure
T	300 K	Gas temperature
NACA profile	0012	Symmetric, 12% thickness
α	$0°$	Angle of attack
d_{wire}	0.2 mm	Wire grid wire diameter
s_{wire}	2 mm	Wire grid spacing (centre-to-centre)
ΔD	0.10 m	Vessel inner diameter surplus over propeller

1.2 Geometry and Coordinate System

The blade rotates in the xy-plane. The z-axis points upward from the blade centre-plane. The NACA 0012 blade has maximum thickness $t_{\max} = 0.12\,c$, so the upper blade surface is at $z_{\text{blade}} =$

$t_{\max}/2 = 0.06\,c$ above the rotation axis. A horizontal wire grid sits at height z_{grid} above the blade top (in this analysis we evaluate conditions at an arbitrary z_{obs}). The cylindrical vessel has inner radius

$$R_{\text{cyl}} = R + \frac{\Delta D}{2}\,. \tag{1}$$

The tip clearance is $\Delta D/2$.

2 Xenon Gas Properties

Xenon is a monatomic ideal gas to good approximation at 15 bar, 300 K (compressibility factor $Z \approx 0.98$).

Density.

$$\rho = \frac{P M_{\text{Xe}}}{R_g T}\,, \tag{2}$$

where $M_{\text{Xe}} = 0.131\,29\,\text{kg/mol}$ and $R_g = 8.314\,\text{J/(mol}\cdot\text{K)}$. This gives $\rho \approx 79\,\text{kg/m}^3$, roughly 65 times denser than air at STP.

Dynamic viscosity. The viscosity of Xe is nearly pressure-independent at moderate densities. At $T = 300\,\text{K}$:

$$\mu \approx 2.32 \times 10^{-5}\,\text{Pa}\cdot\text{s}\,. \tag{3}$$

Kinematic viscosity.

$$\nu = \frac{\mu}{\rho} \approx 2.94 \times 10^{-7}\,\text{m}^2/\text{s}\,. \tag{4}$$

This is extremely small — about 50 times smaller than air at STP — because ρ is large while μ is comparable to air.

Speed of sound. For a monatomic ideal gas with $\gamma = 5/3$:

$$c_s = \sqrt{\frac{\gamma R_g T}{M_{\text{Xe}}}} \approx 178\,\text{m/s}\,. \tag{5}$$

3 Blade Kinematics and Flow Regime

3.1 Bang-Bang Motion Profile with Aerodynamic Drag

The propeller executes a *bang-bang* motion profile: it accelerates under constant motor torque τ, then decelerates under reversed torque $-\tau$. The motion is governed by:

$$I\,\frac{d\omega}{dt} = \pm\tau - \tau_{\text{drag}}(\omega)\,, \tag{6}$$

where the upper sign applies during acceleration and the lower during deceleration.

Moment of Inertia. The system consists of two blades (uniform rods of mass m_{blade} extending from $r = 0$ to $r = R$) and two carrier plates (point masses m_{plate} at the worst-case position $r = R$):

$$I = 2 \times \frac{1}{3}\,m_{\text{blade}}\,R^2 + 2 \times m_{\text{plate}}\,R^2 = \frac{2R^2}{3}\,(m_{\text{blade}} + 3\,m_{\text{plate}})\,. \tag{7}$$

For default parameters ($m_{\text{blade}} = 0.25\,\text{kg}$, $m_{\text{plate}} = 0.15\,\text{kg}$, $R = 1.6\,\text{m}$): $I \approx 1.19\,\text{kg}\cdot\text{m}^2$.

Aerodynamic Drag Torque. The drag on both blades at angular velocity ω is:

$$\tau_{\text{drag}} = \mathcal{D}\,\omega^2\,, \qquad \mathcal{D} = \frac{\rho\, C_d\, d_{\text{wake}}\, R^4}{4}\,, \tag{8}$$

where $d_{\text{wake}} = 0.12\,c$ is the wake thickness (approximately equal to the blade thickness for a streamlined section), and $C_d \approx 0.1$ is an effective drag coefficient for the blade assembly moving through the gas. Note that the NACA 0012 profile drag coefficient at $\alpha = 0$ is much smaller ($C_d^{\text{profile}} \approx 0.007$), but this value applies to flow along the chord. The higher effective C_d used here accounts for: (i) the blade cross-section sweeping tangentially through the gas, and (ii) the carrier plates (m_{plate}) sliding on the blades, which add semi-bluff frontal area and flow disruption at interfaces. The combined blade-plus-plate assembly is thus intermediate between a streamlined airfoil and a bluff body.

For default parameters ($\rho = 79\,\text{kg/m}^3$, $C_d = 0.1$, $d_{\text{wake}} = 0.12 \times 0.16\,\text{m} = 19.2\,\text{mm}$, $R = 1.6\,\text{m}$):

$$\mathcal{D} \approx 0.25\,\text{N}\cdot\text{m}\cdot\text{s}^2/\text{rad}^2\,. \tag{9}$$

Asymmetric Switching Point. Since drag always opposes motion, deceleration (where both reversed motor torque *and* drag oppose motion) is stronger than acceleration. To complete the full sweep angle θ, the switching point θ_{switch} must satisfy $\theta_{\text{switch}} > \theta/2$.

The equation of motion (6) is nonlinear and is solved numerically to find θ_{switch} such that $\omega \to 0$ exactly when $\theta \to \theta_{\text{target}}$.

Inertia-Only Reference (No Drag). For comparison, if drag is neglected:

$$\alpha_0 = \frac{\tau}{I} \approx 50\,\text{rad/s}^2\,, \tag{10}$$

$$t_{\text{rot},0} = 2\sqrt{\frac{\theta\, I}{\tau}} \approx 500\,\text{ms}\,, \tag{11}$$

$$\omega_{\text{max},0} = \sqrt{\frac{\theta\,\tau}{I}} \approx 12.6\,\text{rad/s}\,. \tag{12}$$

Numerical Solution with Drag. For default parameters, the numerical integration yields:

$$\theta_{\text{switch}} \approx 2.05\,\text{rad} \quad (117.5^\circ,\ \text{vs. } 90^\circ \text{ without drag})\,, \tag{13}$$

$$t_{\text{accel}} \approx 306\,\text{ms}\,, \tag{14}$$

$$t_{\text{decel}} \approx 201\,\text{ms}\,, \tag{15}$$

$$t_{\text{rot}} \approx 507\,\text{ms} \quad (+1.4\% \text{ compared to no-drag})\,, \tag{16}$$

$$\omega_{\text{max}} \approx 11.8\,\text{rad/s} \quad (-6.3\% \text{ reduction})\,. \tag{17}$$

Peak Tip Velocity.

$$V_{\text{tip}} = \omega_{\text{max}}\, R \approx 18.8\,\text{m/s}\,. \tag{18}$$

Drag Impact Summary. Aerodynamic drag has a modest effect on the kinematics:

- Rotation time increases by $\sim 1.4\%$.
- Peak angular velocity decreases by $\sim 6.3\%$.
- The switching angle shifts from 90° to 117.5°.

The qualitative behaviour remains unchanged, and the peak velocity is still $\sim 19\,\text{m/s}$, well within the incompressible regime.

3.2 Flow Regime

We evaluate the flow regime at the peak velocity, which represents the most demanding conditions for the boundary layer.

Mach Number. The tip Mach number is

$$\mathrm{Ma}_{\mathrm{tip}} = \frac{V_{\mathrm{tip}}}{c_s}\,. \tag{19}$$

For default parameters, $\mathrm{Ma} \approx 0.11 < 0.3$, so the flow is *incompressible*: density variations are negligible ($< 5\%$), the continuity equation simplifies to $\nabla \cdot \mathbf{v} = 0$, and no shock waves or compressibility-driven phenomena occur.

Reynolds Number. The chord-based Reynolds number at the tip is

$$\mathrm{Re}_{\mathrm{tip}} = \frac{V_{\mathrm{tip}}\, c}{\nu}\,. \tag{20}$$

For default parameters, $\mathrm{Re}_{\mathrm{tip}} \approx 1.1 \times 10^7$, far exceeding the critical value of $\sim 5 \times 10^5$ for transition. The boundary layer is therefore *fully turbulent*: inertial forces dominate viscous forces, producing chaotic eddies with enhanced mixing near the wall. The turbulent BL is thicker than a laminar one but has a fuller velocity profile (more momentum near the wall).

4 Boundary Layer on the Upper Surface

Since we are interested in the flow *above* the blade, only the upper-surface boundary layer (BL) matters.

4.1 Flat-Plate Turbulent BL Estimates

For a turbulent BL developing from the leading edge over a flat plate of length c at Reynolds number $\mathrm{Re}_c = Vc/\nu$, the standard $\frac{1}{7}$-power-law result gives:

$$\delta = 0.37\, c\, \mathrm{Re}_c^{-1/5}\,, \tag{21}$$

$$\theta = \frac{7}{72}\,\delta = \frac{7}{72} \times 0.37\, c\, \mathrm{Re}_c^{-1/5}\,, \tag{22}$$

$$\delta^* = \frac{1}{8}\,\delta\,, \tag{23}$$

where δ is the BL thickness, θ the momentum thickness, and δ^* the displacement thickness.

4.2 NACA 0012 Pressure Gradient Effects

An explicit integral BL calculation (Thwaites' method for the laminar portion, Head's entrainment method for the turbulent portion) using the NACA 0012 inviscid velocity distribution shows that the favourable pressure gradient over the front $\sim 30\%$ of the chord thins the BL, while the adverse gradient over the rear $\sim 70\%$ thickens it. These two effects approximately cancel: at $x/c \approx 0.95$ (the last station before the integral method approaches separation), $\delta \approx 4.0\,\mathrm{mm}$, in excellent agreement with the flat-plate estimate of $3.9\,\mathrm{mm}$. We therefore use the flat-plate formula (21) throughout.

4.3 Skin Friction and Drag Coefficient

The local skin-friction coefficient from the Blasius correlation:

$$C_f = 0.0592\,\mathrm{Re}_c^{-1/5}\,. \tag{24}$$

The friction velocity:

$$u_\tau = V_{\mathrm{tip}}\sqrt{\frac{C_f}{2}}\,. \tag{25}$$

The profile drag coefficient for NACA 0012 at $\alpha = 0$ is obtained from the Squire–Young formula applied at the trailing edge, or from experimental data:

$$C_d \approx 0.006\text{–}0.009 \tag{26}$$

depending on the transition location. For the numerics we compute C_d from the BL quantities via

$$C_d = \frac{2\,(2\theta)}{c}\left(\frac{U_e(x_{\mathrm{TE}})}{V}\right)^{(H+5)/2}, \tag{27}$$

where $H = \delta^*/\theta$ is the shape factor at the trailing edge.

4.4 Wake Geometry (Upper Surface Only)

The relevant length scales for default parameters are:

$$\delta \approx 2.3\,\mathrm{mm}\,, \quad \theta \approx 0.23\,\mathrm{mm}\,, \quad \delta^* \approx 0.29\,\mathrm{mm}\,. \tag{28}$$

The upper blade surface sits at $z_{\mathrm{blade}} = 0.06\,c$ above the rotation axis. The BL extends to

$$z_{\mathrm{BL}} = 0.06\,c + \delta\,. \tag{29}$$

The gap between the top of the BL and the observation height is

$$\Delta z_{\mathrm{gap}} = z_{\mathrm{obs}} - \delta\,. \tag{30}$$

For default parameters ($z_{\mathrm{obs}} = 5\,\mathrm{mm}$, $\delta \approx 2.3\,\mathrm{mm}$), $\Delta z_{\mathrm{gap}} \approx 2.7\,\mathrm{mm}$.

4.5 Initial Velocity Profile in the Lab Frame

In the blade's reference frame, the BL velocity follows a $\frac{1}{7}$-power law: $u_{\mathrm{blade}}(z) = V(z/\delta)^{1/7}$ for $0 \le z \le \delta$, with $u = V$ outside the BL. Transforming to the lab frame (where the gas is initially at rest and the blade moves at V):

$$u_{\mathrm{lab}}(z) = \begin{cases} V\left[1 - (z/\delta)^{1/7}\right] & 0 < z \le \delta \\ 0 & z > \delta \end{cases} \tag{31}$$

The gas near the blade surface ($z \to 0$) is dragged along at nearly V; at $z = \delta$ the velocity is zero.

5 Mechanism 1: Turbulent Eddy Diffusion

5.1 Statement of the Problem

After the blade stops, the momentum in the BL (equation 31) can in principle diffuse vertically into the quiescent gas above. The question is whether this diffusion carries any appreciable velocity to $z = z_{\mathrm{obs}}$ within time t_{obs}.

The evolution is governed by the one-dimensional diffusion equation:

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial z}\left[\nu_{\text{eff}}(z,t)\,\frac{\partial u}{\partial z}\right], \tag{32}$$

with initial condition (31) and boundary conditions

$$u(0,t) = 0 \quad \text{(no-slip on stationary blade)}\,, \qquad u(z\to\infty,t) = 0\,. \tag{33}$$

5.2 Turbulent Viscosity Profile

The effective viscosity is $\nu_{\text{eff}} = \nu + \nu_t(z,t)$. At $t = 0$ (blade just stopped), the turbulent viscosity in the BL follows the standard two-layer model:

$$\nu_t(z,0) = \begin{cases} \kappa\, u_\tau\, z & z < 0.2\,\delta \quad \text{(inner / log-law region)} \\ 0.018\, V\, \delta^* & 0.2\,\delta \le z < \delta \quad \text{(outer region)} \\ 0 & z \ge \delta \end{cases} \tag{34}$$

where $\kappa = 0.41$ is the von Kármán constant. For default parameters:

$$\nu_t^{\text{inner}}(z = 0.2\delta) = \kappa\, u_\tau\,(0.2\delta) \approx 1.3\times 10^{-4}\,\text{m}^2/\text{s}\,, \tag{35}$$

$$\nu_t^{\text{outer}} = 0.018\, V\, \delta^* \approx 1.0\times 10^{-4}\,\text{m}^2/\text{s}\,. \tag{36}$$

The ratio to molecular viscosity is $\nu_t/\nu \approx 350$.

5.3 Turbulence Decay After Blade Stops

Once the blade stops, there is no energy input to sustain the turbulence. The eddies decay on the large-eddy turnover timescale:

$$t_e = \frac{\delta}{u'}\,, \tag{37}$$

where $u' \approx 0.05\,V$ is the outer-layer turbulence intensity (comparable to u_τ). For default parameters, $t_e \approx 2.3\,\text{ms}$.

We model the decay as exponential with an e-folding time of $2\,t_e$ (approximately two eddy turnovers for significant energy loss):

$$\nu_t(z,t) = \nu_t(z,0)\,\exp\!\left(-\frac{t}{2\,t_e}\right). \tag{38}$$

5.4 Diffusion Length Scale

In the brief window before the turbulence dies ($\sim 2\,t_e \approx 5\,\text{ms}$), the turbulent diffusion extends momentum by a distance

$$\ell_{\text{turb}} \sim \sqrt{\nu_t^{\text{outer}} \times 2\,t_e} \approx \sqrt{1.0\times 10^{-4}\times 0.005} \approx 0.7\,\text{mm}\,. \tag{39}$$

Starting from the top of the BL at $z = \delta \approx 2.3\,\text{mm}$, the momentum can reach at most $z \approx 3.0\,\text{mm}$. For $z_{\text{obs}} = 5\,\text{mm}$, the gap of $2.7\,\text{mm}$ is larger than ℓ_{turb}.

After the turbulence decays, only molecular diffusion remains. In the remaining time ($\sim 0.5\,\text{s}$):

$$\ell_{\text{mol}} = \sqrt{\nu\, t_{\text{obs}}} \approx \sqrt{2.94\times 10^{-7}\times 0.5} \approx 0.38\,\text{mm}\,. \tag{40}$$

The total reach is $\delta + \ell_{\text{turb}} + \ell_{\text{mol}} \approx 3.4\,\text{mm}$, short of $z_{\text{obs}} = 5\,\text{mm}$.

5.5 No-Slip Absorption

Simultaneously, the no-slip condition at $z = 0$ (the now-stationary blade surface) absorbs momentum from below. Since most of the momentum is concentrated near $z = 0$ (equation 31), this absorption is efficient. The numerical solution of equation (32) shows that $\sim 38\%$ of the initial momentum is absorbed by the surface within 0.5 s.

5.6 Result

The numerical solution of the 1D diffusion equation (32) with decaying turbulent viscosity gives:

$$\boxed{u(z_{\rm obs},\, t_{\rm obs}) \approx 0 \quad \text{(eddy diffusion)}} \tag{41}$$

The turbulent eddies are too small ($\sim \delta = 2.3\,$mm) and decay too quickly ($\sim 5\,$ms) to transport momentum across the gap to $z_{\rm obs}$.

6 Mechanism 2: Bulk Swirl and Angular Momentum

6.1 Angular Impulse

The drag on both blades deposits tangential (azimuthal) momentum into the gas. The drag per unit span at radius r is

$$f(r) = \tfrac{1}{2}\,\rho\,[\omega r]^2\,c\,C_d\,. \tag{42}$$

The total torque:

$$\tau = 2\int_0^R r\,f(r)\,dr = \rho\,\omega^2\,c\,C_d\,\frac{R^4}{4}\,. \tag{43}$$

The angular impulse over the sweep:

$$J = \tau\,T_{\rm sweep}\,. \tag{44}$$

6.2 Where the Momentum Resides

This angular momentum is deposited *exclusively* within the boundary layers on the blade surfaces — a slab of thickness $\sim (t_{\rm max} + 2\delta)$ centred on the blade plane. The gas outside this slab is undisturbed (apart from the potential flow, which vanishes when the blade stops; see Section 7).

The key question is whether this momentum can propagate vertically to $z_{\rm obs}$.

6.3 Molecular Diffusion Timescale

The time for momentum to diffuse across the gap $\Delta z_{\rm gap}$ by molecular viscosity alone:

$$t_{\rm visc} = \frac{(\Delta z_{\rm gap})^2}{\nu} = \frac{(z_{\rm obs} - \delta)^2}{\nu}\,. \tag{45}$$

For default parameters, $t_{\rm visc} = (0.0027)^2/2.94\times10^{-7} \approx 25\,$s — much longer than $t_{\rm obs} = 0.5\,$s.

6.4 Ekman Pumping

In a rotating fluid adjacent to a boundary, the Ekman layer drives a secondary (radial then axial) circulation. The Ekman layer thickness and the resulting axial velocity are:

$$\delta_E = \sqrt{\frac{\nu_{\rm eff}}{\Omega}}\,, \qquad w_E = \sqrt{\nu_{\rm eff}\,\Omega}\,, \tag{46}$$

where $\Omega \sim u_{\rm wake}/R$ is the effective rotation rate of the wake gas.

With molecular viscosity: $\delta_E \approx 0.22\,\mathrm{mm}$, $w_E \approx 1.4\,\mathrm{mm/s}$. Negligible.

With turbulent viscosity ($\nu_t \approx 1 \times 10^{-4}\,\mathrm{m^2/s}$): $\delta_E \approx 4\,\mathrm{mm}$, $w_E \approx 2.6\,\mathrm{cm/s}$. But the turbulence decays in $\sim 5\,\mathrm{ms}$, so the vertical distance traversed is $w_E \times 2t_e \approx 0.12\,\mathrm{mm}$.

In both cases, Ekman pumping cannot reach z_{obs} from the blade plane.

6.5 Ekman Spin-Up Time

Even if one posits a mechanism for establishing vessel-scale secondary circulation, the Ekman spin-up timescale is:

$$t_{\mathrm{Ekman}} = \frac{H}{\sqrt{\nu_{\mathrm{eff}}\,\Omega}}\,, \tag{47}$$

where H is the vessel height. With turbulent ν: $t_{\mathrm{Ekman}} \approx 39\,\mathrm{s}$. With molecular ν: $t_{\mathrm{Ekman}} \approx 740\,\mathrm{s}$. Both are far longer than t_{obs}.

6.6 Result

$$\boxed{u(z_{\mathrm{obs}},\, t_{\mathrm{obs}}) \approx 0 \quad \text{(bulk swirl)}} \tag{48}$$

The angular momentum stays confined to the blade plane. No mechanism transports it to z_{obs} within t_{obs}.

7 Mechanism 3: Displacement (Potential) Flow

7.1 Flow Field During the Sweep

The NACA 0012 blade has finite thickness $t_{\mathrm{max}} = 0.12\,c$. As it moves through the gas, it displaces fluid. The inviscid (potential) flow field around a two-dimensional body of effective diameter $d \approx t_{\mathrm{max}}$ moving at velocity V produces a vertical velocity component at height z above the blade centre that scales as:

$$w_{\mathrm{disp}}(z) \sim V\,\frac{(d/2)^2}{z^2 + (d/2)^2}\,. \tag{49}$$

For $z = z_{\mathrm{obs}}$ and default parameters, this gives $w_{\mathrm{disp}} \sim 16\,\mathrm{m/s}$ — a large value, present *only while the blade is in motion and in the vicinity.*

7.2 Acoustic Clearing Time

The potential flow field is slaved to the blade. When the blade stops, the disturbance propagates away at the speed of sound:

$$t_{\mathrm{acoustic}} = \frac{R}{c_s} \approx 9\,\mathrm{ms}\,. \tag{50}$$

Within $\sim 10\,\mathrm{ms}$ of the blade stopping, the potential flow field has been radiated away as a weak acoustic wave.

7.3 Result

$$\boxed{u(z_{\mathrm{obs}},\, t_{\mathrm{obs}}) = 0 \quad \text{(potential flow, for } t_{\mathrm{obs}} \gg t_{\mathrm{acoustic}})} \tag{51}$$

8 Mechanism 4: Pressure Pulse from Deceleration

When the blade decelerates to rest over some time t_{decel}, the change in momentum of the fluid near the blade creates a pressure transient.

8.1 Amplitude

The pressure perturbation scales as:

$$\Delta p \sim \rho\, V_{\text{tip}}^2 \,. \tag{52}$$

For default parameters, $\Delta p \approx 32\,000\,\text{Pa}$, which is 2% of the ambient 15 bar — a small perturbation.

The velocity induced by this acoustic wave:

$$\Delta u \sim \frac{\Delta p}{\rho\, c_s} = \frac{V_{\text{tip}}^2}{c_s} \,. \tag{53}$$

For default parameters, $\Delta u \approx 2.3\,\text{m/s}$.

8.2 Propagation

The pressure pulse arrives at z_{obs} in a time

$$t_{\text{arrive}} = \frac{z_{\text{obs}}}{c_s} \,. \tag{54}$$

For default parameters, $t_{\text{arrive}} \approx 28\,\mu\text{s}$. The pulse has duration $\sim t_{\text{decel}}$ and then propagates past z_{obs}, leaving no residual velocity.

8.3 Result

$$\boxed{u(z_{\text{obs}},\, t_{\text{obs}}) = 0 \quad (\text{pressure pulse, for } t_{\text{obs}} \gg t_{\text{decel}})} \tag{55}$$

9 Mechanism 5: Secondary Flows

9.1 Tip Clearance Flow

In turbomachinery, a pressure difference between the suction and pressure sides of a blade drives leakage flow through the tip gap. For a lifting blade this is a significant source of axial mixing.

For NACA 0012 at $\alpha = 0$: there is *no lift*, and hence *no pressure difference* across the blade. The flow field is symmetric about the chord plane.

$$\Delta p_{\text{tip}} = 0 \qquad (\alpha = 0) \,. \tag{56}$$

No tip clearance flow exists.

9.2 Centrifugal Effects and Wake–Wall Interaction

The tangential wake gas experiences centrifugal acceleration:

$$a_{\text{cent}} = \frac{u_{\text{wake}}^2}{R} \,. \tag{57}$$

The time for gas to traverse the tip gap $\Delta r = \Delta D/2$ under this acceleration:

$$t_{\text{cent}} = \sqrt{\frac{2\,\Delta r}{a_{\text{cent}}}} = \sqrt{\frac{2\,\Delta r\, R}{u_{\text{wake}}^2}} \,. \tag{58}$$

For default parameters, $t_{\text{cent}} \approx 0.04\,\text{s}$, but the "rotation" that drives the centrifugal force is itself decaying. Furthermore, the wake is *tangential*: it flows *along* the cylindrical wall, not *into* it. There is no face-on impingement that would deflect the flow into the axial direction.

9.3 Axial Coupling Factor "ε"

One might posit an azimuthal-to-axial coupling fraction ε such that $v_{\text{axial}} \sim \varepsilon V_{\text{mean}}$. This parameter has *no basis* in any identifiable flow mechanism at $\alpha = 0$. All candidate coupling mechanisms (Ekman pumping, tip clearance, centrifugal wall deflection) have been shown to be either absent or far too slow.

9.4 Result

$$\boxed{u(z_{\text{obs}},\, t_{\text{obs}}) = 0 \quad \text{(secondary flows)}} \tag{59}$$

10 Mechanism 6: Vortex Shedding

When the blade stops, any bound vorticity is shed. For a *non-lifting* blade ($\alpha = 0$), there is no net circulation, and hence no starting/stopping vortex of macroscopic strength.

Drag-induced vortex shedding (von Kármán street) is characterised by the Strouhal number:

$$\text{St} = \frac{f\, t_{\text{max}}}{V}\,, \tag{60}$$

where St ≈ 0.21 for bluff bodies. But NACA 0012 at $\alpha = 0$ is *streamlined*: the wake is thin and coherent vortex shedding is strongly suppressed. Any residual shed vortices are confined to the blade plane (within $\sim \delta$ of the surface) and subject to the same vertical transport limitations established in Section 5.

$$\boxed{u(z_{\text{obs}},\, t_{\text{obs}}) = 0 \quad \text{(vortex shedding)}} \tag{61}$$

11 Conclusion

Table 3 summarises all six mechanisms. Every one yields zero residual velocity at z_{obs} for $t_{\text{obs}} \gg t_{\text{acoustic}}$.

Table 3: Summary of all mechanisms for residual velocity at z_{obs} above the blade, t_{obs} after stopping. Default parameters: $c = 0.16\,\text{m}$, $\theta_{\text{sweep}} = \pi$, $t_{\text{rot}} = 507\,\text{ms}$ (with drag), $z_{\text{obs}} = 5\,\text{mm}$, $t_{\text{obs}} = 0.5\,\text{s}$.

Mechanism	Characteristic timescale	Reach / Comment	$u(z_{\text{obs}}, t_{\text{obs}})$
Eddy diffusion	$2t_e \approx 5\,\text{ms}$	$\delta + 0.7\,\text{mm} = 3.0\,\text{mm}$	0
Bulk swirl	$t_{\text{visc}} \approx 25\,\text{s}$	Momentum stays in blade plane	0
Potential flow	$t_{\text{acoustic}} \approx 9\,\text{ms}$	Vanishes at speed of sound	0
Pressure pulse	$t_{\text{decel}} \sim 10\,\text{ms}$	Transient wave, passes through	0
Secondary flows	$t_{\text{Ekman}} \sim 39\,\text{s}$	No lift $\Rightarrow$ no tip clearance	0
Vortex shedding	—	Confined to blade plane	0

The gas at z_{obs} above the blade surface is **quiescent** at time t_{obs} after the blade stops, for any t_{obs} much longer than the acoustic clearing time ($\sim 10\,\text{ms}$) and for any z_{obs} larger than the boundary layer thickness ($\sim 2.3\,\text{mm}$ for default parameters).

The fundamental reason is that the NACA 0012 at zero angle of attack produces no lift, so all aerodynamic effects are confined to the thin drag wake within δ of the blade surface. The dense xenon gas has an extremely small kinematic viscosity, making all diffusion timescales very long, while the turbulent eddies are small ($\sim \delta$) and decay rapidly ($\sim 5\,\text{ms}$) once the blade stops.